

Development and validation of an ultra-high performance liquid chromatography-tandem mass spectrometry method for quantifying lenacapavir plasma concentrations: Application to therapeutic monitoring.

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Although current antiretroviral therapy (ART) effectively suppresses HIV in the blood, regimens may fail due to suboptimal treatment history and non-adherence to ART. In these scenarios, accumulation of viral resistance mutations to ART drug classes may occur. For these treatment-experienced people living with HIV (PLWH), activity against resistant viral strains is required; lack of therapeutic efficacy will result in continued viral replication and progression to acquired immunodeficiency syndrome. New treatment options have emerged. Lenacapavir is a first-in-class long-acting HIV-1 capsid inhibitor approved for the treatment of HIV in treatment-experienced patients. Lenacapavir is approved with an initiation regimen of oral and subcutaneous injection dosing followed by subcutaneous self-injection every 6 months. With infrequent dosing, therapeutic drug monitoring may be necessary to ensure adequate concentrations are consistently achieved in the plasma to assure treatment adherence and prevent further HIV resistance formation. To this end, we developed and validated a highly selective ultra-high performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) method to quantify lenacapavir concentrations in human plasma. A simple protein precipitation with acetonitrile followed by supernatant dilution was performed. Lenacapavir and its stable labeled internal standard were separated at 1.90 min using a multi-step UPLC gradient. The assay for lenacapavir quantification was extensively validated according to the United States Food and Drug Administration Bioanalytical Guidelines over a clinically relevant range of 0.1 to 500 ng/mL with excellent linearity ($R^2 \geq 0.9960$). This analytical method achieves acceptable performance of trueness (89.7-104.1 %), repeatability, and precision ($CV < 15\%$). We applied this method to quantify a clinical sample and to determine the percent protein-unbound. This method can be utilized for the therapeutic monitoring of lenacapavir in human plasma for monitoring HIV treatment efficacy.

Efficacy and safety of the novel capsid inhibitor lenacapavir to treat multidrug-resistant HIV: week 52 results of a phase 2/3 trial.

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Lenacapavir, a first-in-class HIV-1 capsid inhibitor, is in development as a long-acting agent for treating and preventing HIV-1. We aimed to evaluate the efficacy and safety of lenacapavir with an optimised background regimen in adults living with multidrug-resistant HIV-1 up to 52 weeks. This ongoing, international, phase 2/3 trial at 42 sites included adults living with multidrug-resistant HIV-1. In cohort 1, 36 participants were randomly assigned (2:1) to add oral lenacapavir (600 mg, days 1 and 2; 300 mg, day 8) or placebo to an existing failing regimen. At day 15, those on oral lenacapavir received subcutaneous lenacapavir 927 mg every 26 weeks; those on placebo started lenacapavir (2-week oral lead-in then subcutaneous). Cohort 1 started an optimised background regimen on day 15. In cohort 2 (non-randomised), 36 participants started an optimised background regimen concurrent with lenacapavir (oral to subcutaneous). Here we report the secondary endpoints of plasma HIV-1 RNA of less than 50 copies per mL or less than 200 copies per mL at week 52 (US Food and Drug Administration snapshot algorithm) in cohort 1 along with results for cohorts 1 and 2 combined. This trial is registered with ClinicalTrials.gov, NCT04150068, and clinicaltrialregister.eu, EudraCT 2019-003814-16 and is ongoing. Of 72 participants, 46 (64%) had CD4 counts of less than 200 cells per μ L and 38 (53%) had no more than

one fully active antiretroviral drug at baseline. In cohort 1, 30 of 36 participants (83%, 95% CI 67-94) had less than 50 HIV-1 RNA copies per mL and 31 of 36 participants (86%, 71-95) had less than 200 HIV RNA copies per mL, at week 52. In all, nine participants (four in cohort 1, five in cohort 2) had emergent lenacapavir resistance; four resuppressed (HIV-1 RNA <50 copies per mL) while maintaining lenacapavir use. One participant discontinued study drug owing to injection site reaction. In participants with multidrug-resistant HIV-1, subcutaneous lenacapavir in combination with an optimised background regimen resulted in a high rate of virological suppression up to 52 weeks. Gilead Sciences.

Highlights on the Development, Related Patents, and Prospects of Lenacapavir: The First-in-Class HIV-1 Capsid Inhibitor for the Treatment of Multi-Drug-Resistant HIV-1 Infection.

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The multidrug-resistant (MDR) human immunodeficiency virus 1 (HIV-1) infection is an unmet medical need. HIV-1 capsid plays an important role at different stages of the HIV-1 replication cycle and is an attractive drug target for developing therapies against MDR HIV-1 infection. Lenacapavir (LEN) is the first-in-class HIV-1 capsid inhibitor approved by the USFDA, EMA, and Health Canada for treating MDR HIV-1 infection. This article highlights the development, pharmaceutical aspects, clinical studies, patent literature, and future directions on LEN-based therapies. The literature for this review was collected from PubMed, authentic websites (USFDA, EMA, Health Canada, Gilead, and NIH), and the free patent database (Espacenet, USPTO, and Patent scope). LEN has been developed by Gilead and is marketed as Sunlenca (tablet and subcutaneous injection). The long-acting and patient-compliant LEN demonstrated a low level of drug-related mutations, is active against MDR HIV-1 infection, and does not reveal cross-resistance to other anti-HIV drugs. LEN is also an excellent drug for patients having difficult or limited access to healthcare facilities. The literature has established additive/synergistic effects of combining LEN with rilpivirine, cabotegravir, islatravir, bictegravir, and tenofovir. HIV-1 infection may be accompanied by opportunistic infections such as tuberculosis (TB). The associated diseases make HIV treatment complex and warrant drug interaction studies (drug-drug, drug-food, and drug-disease interaction). Many inventions on different aspects of LEN have been claimed in patent literature. However, there is a great scope for developing more inventions related to the drug combination of LEN with anti-HIV/anti-TB drugs in a single dosage form, new formulations, and methods of treating HIV and TB co-infection. Additional research may provide more LEN-based treatments with favorable pharmacokinetic parameters for MDR HIV-1 infections and associated opportunistic infections such as TB.

Lenacapavir administered every 26 weeks or daily in combination with oral daily antiretroviral therapy for initial treatment of HIV: a randomised, open-label, active-controlled, phase 2 trial.

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Antiretroviral agents with novel mechanisms and dosing intervals could expand treatment options for people with HIV. Lenacapavir, an inhibitor of capsid protein that makes use of a unique mechanism, can be administered orally or subcutaneously. We sought to explore the efficacy of lenacapavir in various combination regimens as initial and maintenance therapy for HIV. In a phase 2, randomised, open-label,

ongoing study at 41 investigational sites in the USA and Dominican Republic, we randomly assigned adults with HIV who had not previously received antiretrovirals to four groups (2:2:2:1). Randomisation was stratified by plasma HIV-1 RNA load ($\leq 100\,000$ or $>100\,000$ copies per mL) at screening. Groups 1 and 2 both received lenacapavir (927 mg) subcutaneously every 26 weeks (after 2 weeks of oral loading [600 mg on days 1 and 2, followed by 300 mg on day 8]) with oral daily emtricitabine (200 mg) and tenofovir alafenamide (25 mg) for 28 weeks followed by subcutaneous lenacapavir (927 mg) plus oral daily tenofovir alafenamide (25 mg, group 1) or bictegravir (75 mg, group 2). Group 3 received oral daily lenacapavir (600 mg on days 1 and 2, followed by 50 mg daily) with emtricitabine (200 mg) and tenofovir alafenamide (25 mg). Group 4 received oral daily bictegravir (50 mg), emtricitabine (200 mg), and tenofovir alafenamide (25 mg). Participants and investigators were not masked to group assignment. The primary endpoint was the percentage of participants with virological suppression (HIV-1 RNA <50 copies per mL) at week 54, analysed in the full analysis set (all randomly assigned participants who received at least one dose of study drug) using only on-treatment data. The safety outcome measures were incidences of treatment-emergent adverse events and graded laboratory abnormalities, analysed in the full analysis set. This study is registered at ClinicalTrials.gov, NCT04143594. Between Nov 22, 2019, and Aug 27, 2020, 249 people with HIV were screened, 183 participants were randomly assigned and 182 received a dose of antiretroviral drugs (52 in group 1, 53 in group 2, 52 in group 3, and 25 in group 4). 22 participants did not complete the full study course (five in group 1, 12 in group 2, four in group 3, and one in group 4). At week 54, virological suppression was 90% (47 of 52 patients) for group 1 (difference vs group 4: -2.6%, 95% CI -18.4 to 13.2), 85% (45 of 53) for group 2 (-7.1%, -23.4 to 9.3), 85% (44 of 52) for group 3 (-7.2%, -23.5 to 9.1), and 92% (23 of 25) for group 4. The most frequent non-injection-site adverse events with lenacapavir (subcutaneous or oral) were headache (13%, 21 of 157) and nausea (13%, 21 of 157). The most common lenacapavir-related injection-site reactions were erythema (27%, 28 of 105), swelling (23%, 24 of 105), and pain (19%, 20 of 105), which were generally mild or moderate. No serious adverse event related to study treatment occurred. Three participants discontinued subcutaneous lenacapavir because of grade 1 injection-site reactions (two for induration and one for erythema or swelling). Lenacapavir warrants further investigation as a potential antiretroviral used orally and as injection in combination with other antiretroviral drugs. Gilead Sciences.

Indirect Treatment Comparisons of Lenacapavir Plus Optimized Background Regimen Versus Other Treatments for Multidrug-Resistant Human Immunodeficiency Virus.

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Heavily treatment-experienced (HTE) people with human immunodeficiency virus (HIV) (PWH) may not achieve virologic suppression (VS) with combination antiretroviral therapy due to multidrug resistance (MDR), intolerance, and safety concerns. These PWH often receive highly individualized treatment regimens, but these regimens may not enable PWH to achieve VS, thereby halting disease progression. Novel medications are required for treating individuals with MDR HIV. Lenacapavir (LEN), a first-in-class HIV capsid inhibitor, is under investigation for the treatment of HTE individuals with MDR HIV in the phase 2/3 CAPELLA study. This study aimed to compare LEN plus optimized background regimen (OBR) with fostemsavir (FTR) + OBR, ibalizumab (IBA) + OBR, and OBR alone in terms of VS, CD4 cell count change from baseline, immunologic recovery, and discontinuation due to adverse events, using indirect treatment comparisons. A systematic review identified clinical evidence on HIV-1 treatments in HTE PWH. A feasibility assessment evaluated the identified studies for indirect treatment comparison analyses based on population characteristics, interventions, comparators, and outcomes of interest. Unanchored simulated treatment comparisons of LEN + OBR versus comparators were conducted. LEN + OBR had

6.57 times higher odds of VS at weeks 24 to 28 than FTR + OBR (95% confidence interval [CI] 1.34-32.28), 8.93 times higher odds of VS than IBA + OBR (95% CI 2.07-38.46), and 12.74 times higher odds of VS than OBR alone (95% CI 1.70-95.37). Change from baseline in CD4 cell count was similar across LEN + OBR, FTR + OBR, and IBA + OBR. LEN + OBR has statistically significantly greater odds of VS at weeks 24 to 28 than its comparators and represents a novel treatment for people with MDR HIV.

Rapid Selection of HIV-2 Capsid Mutations in Salvage Therapy With Lenacapavir-containing Regimen.

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Lenacapavir is the first capsid inhibitor and its use is currently approved for multidrug-resistant human immunodeficiency virus-1 infection. We report that, despite the initial efficacy of a lenacapavir-containing regimen in patients with multidrug-resistant human immunodeficiency virus-2 viruses, virological suppression was not achieved after a year and most patients selected capsid drug resistance-associated mutations.

Efficacy and Safety of Switching to Daily Bictegravir Plus Lenacapavir From a Complex HIV Treatment Regimen: A Randomized, Open-Label, Multicenter Phase 2 Study (ARTISTRY-1).

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Complex antiretroviral therapy (ART) regimens, such as those requiring multiple tablets, several doses per day, or both, can negatively affect quality of life and treatment adherence among people with human immunodeficiency virus (HIV). ARTISTRY-1 is a phase 2/3, operationally seamless, randomized, open-label, multicenter, active-controlled study (GS-US-621-6289; NCT05502341). Phase 2 of the study enrolled adults with plasma HIV-1 RNA <50 copies/mL receiving a complex ART regimen for ≥ 6 months. Efficacy and safety outcomes were evaluated after a switch to bictegravir (BIC) (75-mg) + lenacapavir (LEN) (25- or 50-mg) regimens, compared with continuing on a complex ART regimen through 24 weeks. Overall, 128 participants were assigned randomly to begin BIC 75 mg + LEN 25 mg ($n = 51$) or BIC 75 mg + LEN 50 mg ($n = 52$) or continue on their complex ART regimen ($n = 25$). At week 24, HIV-1 RNA was ≥ 50 copies/mL in 0 of 51, 1 of 52 (1.9%), and 0 of 25 participants in the 3 groups, respectively. CD4 cell counts and percentages remained stable through week 24; the median change from baseline in CD4 cell count (interquartile range) was 18 (-39 to 70), -16 (-80 to 93), and 42 (-36 to 90) cells/ μ L, respectively. There were no study discontinuations due to a serious adverse event through week 24. Both BIC + LEN dosing regimens were well tolerated, with similar safety profiles observed between groups. These data support the continued evaluation of the combination of BIC and LEN to optimize treatment in people with HIV and virologic suppression who are receiving complex ART regimens.

Absence of Lenacapavir (GS-6207) Phenotypic Resistance in HIV Gag Cleavage Site Mutants and in Isolates with Resistance to Existing Drug Classes.

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Lenacapavir (LEN; GS-6207) is a potent first-in-class inhibitor of HIV-1 capsid with long-acting properties and the potential for subcutaneous dosing every 3 months or longer. In the clinic, a single subcutaneous LEN injection (20 mg to 750 mg) in people with HIV (PWH) induced a strong antiviral response, with a >2.3 mean log₁₀ decrease in HIV-1 RNA at day 10. HIV-1 Gag mutations near protease (PR) cleavage sites have emerged with the use of protease inhibitors (PIs). Here, we have characterized the activity of LEN in mutants with Gag cleavage site mutations (GCSMs) and mutants resistant to other drug classes. HIV mutations were inserted into the pXXLAI clone, and the resulting mutants ($n = 70$) were evaluated using a 5-day antiviral assay. LEN EC₅₀ fold change versus the wild type ranged from 0.4 to 1.9 in these mutants, similar to that for the control drug. In contrast, reduced susceptibility to PIs and maturation inhibitors (MIs) was observed. Testing of isolates with resistance against the 4 main classes of drugs ($n = 40$) indicated wild-type susceptibility to LEN (fold change ranging from 0.3 to 1.1), while reduced susceptibility was observed for control drugs. HIV GCSMs did not impact the activity of LEN, while some conferred resistance to MIs and PIs. Similarly, LEN activity was not affected by naturally occurring variations in HIV Gag, in contrast to the reduced susceptibility observed for MIs. Finally, the activity of LEN was not affected by the presence of resistance mutations to the 4 main antiretroviral (ARV) drug classes. These data support the evaluation of LEN in PWH with multiclass resistance.

Pharmacokinetics of long-acting lenacapavir in participants with hepatic or renal impairment.

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Lenacapavir is a novel, first-in-class, multistage inhibitor of HIV-1 capsid function approved for the treatment of multidrug-resistant HIV-1 infection in combination with other antiretrovirals for heavily treatment-experienced people with HIV. Two Phase 1, open-label, parallel-group, single-dose studies assessed the pharmacokinetics (PK) of lenacapavir in participants with moderate hepatic impairment [Child-Pugh-Turcotte (CPT) Class B: score 7-9] or severe renal impairment [$15 \leq$ creatinine clearance (CL_{Cr}) ≤ 29 mL/min] to inform lenacapavir dosing in HIV-1-infected individuals with organ impairment. In both studies, a single oral dose of 300 mg lenacapavir was administered to participants with normal ($n = 10$) or impaired ($n = 10$) hepatic/renal function who were matched for age (± 10 years), sex, and body mass index ($\pm 20\%$). Lenacapavir exposures [area under the plasma concentration-time curve from time 0 to infinity (AUC_{inf}) and maximum concentration (C_{max})] were approximately 1.47- and 2.61-fold higher, respectively, in participants with moderate hepatic impairment compared to those with normal hepatic function, whereas lenacapavir AUC_{inf} and C_{max} were approximately 1.84- and 2.62-fold higher, respectively, in participants with severe renal impairment compared to those with normal renal function. Increased lenacapavir exposures with moderate hepatic or severe renal impairment were not considered clinically meaningful. Lenacapavir was considered generally safe and well tolerated in both studies. These results support the use of approved lenacapavir dosing regimen in patients with mild (CPT Class A: score 5-6) or moderate hepatic impairment as well as in patients with mild ($60 \leq$ CL_{Cr} ≤ 89 mL/min), moderate ($30 \leq$ CL_{Cr} ≤ 59 mL/min), and severe renal impairment.

Efficacy of Integrase Strand Transfer Inhibitors and the Capsid Inhibitor Lenacapavir against HIV-2, and Exploring the Effect of Raltegravir on the Activity of SARS-CoV-2.

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Retroviruses perpetuate their survival by incorporating a copy of their genome into the host cell, a critical step catalyzed by the virally encoded integrase. The viral capsid plays an important role during the viral life cycle, including nuclear importation in the case of lentiviruses and integration targeting events; hence, targeting the integrase and the viral capsid is a favorable therapeutic strategy. While integrase strand transfer inhibitors (INSTIs) are recommended as first-line regimens given their high efficacy and tolerability, lenacapavir is the first capsid inhibitor and the newest addition to the HIV treatment arsenal. These inhibitors are however designed for treatment of HIV-1 infection, and their efficacy against HIV-2 remains widely understudied and inconclusive, supported only by a few limited phenotypic susceptibility studies. We therefore carried out inhibition profiling of a panel of second-generation INSTIs and lenacapavir against HIV-2 in cell culture, utilizing pseudovirion inhibition profiling assays. Our results show that the tested INSTIs and lenacapavir exerted excellent efficacy against ROD-based HIV-2 integrase. We further evaluated the efficacy of raltegravir and other INSTIs against different variants of SARS-CoV-2; however, contrary to previous *in silico* findings, the inhibitors did not demonstrate significant antiviral activity.
